4.6. Unitary representations. Another proof of Maschke's theorem for complex representations

Definition 4.6.1. A **unitary** finite dimensional representation of a group G is a representation of G on a complex finite dimensional vector space V over $\mathbb{C}$ equipped with a G-invariant positive definite Hermitian form[1] $(\ ,\ )$, i.e., such that $\rho_V(g)$ are unitary operators: $(\rho_V(g)v, \rho_V(g)w) = (v, w)$.

Theorem 4.6.2. *If G is finite, then any finite dimensional representation of G has a unitary structure. If the representation is irreducible, this structure is unique up to scaling by a positive real number.*

Proof. Take any positive definite Hermitian form B on V and define another Hermitian form $\mathbf{B}$ on V as follows:

$$\mathbf{B}(v, w) = \sum_{g \in G} B(\rho_V(g)v, \rho_V(g)w).$$

Then $\mathbf{B}$ is a positive definite G-invariant Hermitian form on V.

If V is an irreducible representation and B_1, B_2 are two positive definite G-invariant Hermitian forms on V, then $B_1(v, w) = B_2(Av, w)$ for some linear map $A : V \to V$ (since any positive definite Hermitian form is nondegenerate), and moreover A is also G-invariant, i.e., is a homormphism of representations. Then by Schur's lemma, $A = \lambda \mathrm{Id}$, and clearly $\lambda > 0$. $\square$

Theorem 4.6.2 implies that if V is a finite dimensional representation of a finite group G, then the **complex conjugate representation** $\overline{V}$ (i.e., the same space V with the same addition and the same action of G, but complex conjugate action of scalars) is isomorphic to the dual representation V^*. Indeed, a homomorphism of representations $\overline{V} \to V^*$ is obviously the same thing as an invariant sesquilinear form on V (i.e., a form additive on both arguments which is linear on the first one and antilinear on the second one), and an isomorphism is

[1] Recall that a sesquilinear form on a complex vector space V is an $\mathbb{R}$-bilinear map $(,) : V \times V \to \mathbb{C}$ such that $(zv, w) = (v, \overline{z}w) = z(v, w)$ for $z \in \mathbb{C}$, and a sesquilinear form $(,)$ is Hermitian if $(v, w) = \overline{(w, v)}$. Recall also that a Hermitian form $(,)$ is positive definite if $(v, v) > 0$ for all nonzero $v \in V$.